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COMPLEX NETWORKS

Construction and Analysis of the Global Airport Network Using NetworkX

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LIST OF ABBREVIATIONS

ICAO	International Civil Aviation Organization
IATA	International Air Transport Association
MultiDiGraph	Directed Multigraph
NetworkX	Python Network Analysis Library

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ABSTRACT

This report presents the construction and analysis of a global airport route network using NetworkX, a Python library for the creation and analysis of complex networks. The network is constructed from the OpenFlights dataset, which provides information about airports and airline routes. The study aims to demonstrate how real-world transportation data can be transformed into a graph representation for topological analysis.

Airports are modeled as nodes, while direct flight routes between airports are represented as directed edges. Two graph structures are implemented: a directed multigraph, which preserves individual route records and multiple connections between the same pair of airports, and a directed graph, which represents unique directed connections. Airport records are matched with route endpoints to construct the network, followed by data cleaning to remove self-loops and isolated nodes.

The resulting cleaned network contains 3,214 airports and 66,770 route records in the directed multigraph, corresponding to 36,906 unique directed connections in the directed graph. Network properties are examined using basic topological measures, including network density, average degree, degree centrality, and airport degree rankings. The analysis identifies airports with a large number of direct connections, providing an overview of the network's connectivity structure.

The results demonstrate the application of NetworkX in constructing, processing, and analyzing a real-world complex network. They also provide a foundation for further investigation of structural properties in transportation networks, while acknowledging the limitations of the historical OpenFlights dataset.

CHAPTER 1. INTRODUCTION

1.1 Background

Complex network theory provides a mathematical framework for studying systems composed of interconnected elements. These systems can be found in many real-world domains, including social interactions, biological systems, communication infrastructures, and transportation networks. Despite their differences, such systems can often be represented using a graph consisting of vertices and edges, allowing their structural properties and connectivity patterns to be investigated.

Among these applications, air transportation is a particularly interesting example of a complex network. Airports serve as connection points, while direct flight routes establish relationships between them. The resulting network can be analyzed to identify highly connected airports, examine the distribution of connections, and understand how individual airports contribute to the overall connectivity of the system.

Unlike an undirected network, an airport route network is naturally represented as a directed graph because a flight from one airport to another does not necessarily imply the existence of a flight in the opposite direction. Furthermore, multiple route records may exist between the same pair of airports, corresponding to different airlines or flight services. These characteristics make airport networks a suitable example for demonstrating graph modeling and network analysis.



Figure 1.1 Illustration of the global air transportation network.

The availability of publicly accessible datasets makes it possible to construct and investigate such networks computationally. In this report, the OpenFlights dataset is used as the primary data source. It provides airport information and recorded airline routes, which can be processed and transformed into graph structures using NetworkX, a Python library for the creation, manipulation, and analysis of complex networks.

1.2 Motivation and Objectives

The main motivation of this project is to gain practical experience in applying complex network theory to a real-world transportation system. Rather than treating airport and route information as independent records, the project transforms the available data into a graph representation, making it possible to study the relationships and connectivity patterns within the system.

The specific objectives of this project are as follows:

- To explore and preprocess the OpenFlights airport and route datasets, identifying the information required for network construction.
- To model the airport route system using NetworkX, with airports represented as nodes and direct flight routes represented as directed edges.
- To implement both a directed multigraph and a directed graph in order to preserve individual route records while also examining unique connections between airports.
- To validate and clean the constructed network by identifying and removing self-loops and isolated nodes.
- To examine fundamental topological properties of the resulting network, including network size, density, average degree, degree centrality, and the airports with the highest total degree.
- To provide a visual representation of the constructed network through an interactive visualization interface.

The project focuses on network construction and fundamental topological analysis rather than the operational performance of individual airlines or airports. Therefore, the results are intended to describe the structure of the network represented by the available dataset, rather than provide a complete assessment of real-world air transportation activity.

1.3 Scope and Limitations

This study uses the OpenFlights dataset, which contains airport information and recorded airline routes. The dataset represents a historical snapshot, last updated in June 2014,

and does not necessarily reflect the current global air transportation system.

The analysis is limited to airports and route records that can be matched successfully using the available airport identifiers. Geographical coordinates are used to validate airport locations, while route endpoints are used to establish connections between airports. The constructed network is subsequently cleaned to remove self-loops and isolated nodes.

The analysis primarily considers the topological structure of the network. Factors such as passenger volume, flight frequency, ticket prices, airport capacity, and actual traffic demand are not included in the implemented analysis. Consequently, the number of connections should not be interpreted as a direct measure of an airport's passenger traffic or operational importance.

1.4 Report Structure

This report is organized into six chapters:

- **Chapter 1 – Introduction:** Presents the background, motivation, objectives, scope, and limitations of the project.
- **Chapter 2 – Dataset Description and Network Modeling:** Introduces the OpenFlights dataset and explains how airports and flight routes are represented as graph elements.
- **Chapter 3 – Network Construction and Data Processing:** Describes the data preprocessing workflow, airport matching, graph construction using NetworkX, and network cleaning procedures.
- **Chapter 4 – Network Analysis and Results:** Presents the fundamental topological properties of the constructed network and discusses the results obtained from the analysis.
- **Chapter 5 – Network Visualization:** Introduces the visualization interface and demonstrates how the constructed airport network can be explored interactively.
- **Chapter 6 – Conclusion and Future Work:** Summarizes the completed work, discusses its limitations, and identifies possible directions for further investigation.

CHAPTER 2. DATASET DESCRIPTION AND NETWORK MODELING

2.1 Overview of the OpenFlights Dataset

The OpenFlights dataset is used as the primary data source for constructing the airport route network in this project. It provides information about airports and recorded airline routes, allowing the relationships between airports to be represented as a graph.

The dataset consists of several files, among which two are directly used in this study:

- **Airports dataset (airports.dat.txt):** Contains information about individual airports, including their identifiers, names, geographical locations, and other descriptive attributes.
- **Routes dataset (routes.dat.txt):** Contains records of airline routes, specifying the airline, source airport, and destination airport for each recorded route.

The original dataset used in this project contains 7,698 airport records and 67,663 route records. It represents a historical collection of airline information, with the OpenFlights database last updated in June 2014. Therefore, it should be regarded as a historical dataset rather than a representation of the current global air transportation system.

Airport Data

Each airport record contains several attributes describing an airport. The most relevant fields for this project are summarized in Table 2.1.

Table 2.1 Relevant fields in the OpenFlights airport dataset.

Field	Attribute	Description
1	Airport ID	Unique identifier assigned to each airport.
2	Name	Name of the airport.
3	City	City served by the airport.
4	Country	Country in which the airport is located.
5	IATA	Three-letter IATA airport code.
6	ICAO	Four-letter ICAO airport code.
7	Latitude	Geographic latitude of the airport.
8	Longitude	Geographic longitude of the airport.
9	Altitude	Airport elevation in feet.

Among these attributes, the airport ID is particularly important because it is used as the unique identifier for graph nodes. Airport names, IATA codes, country information, and geographical coordinates are retained as descriptive attributes. The latitude and longitude fields also provide the geographical information required for visualizing airport locations.

Route Data

The route dataset describes direct connections between airports. Each record identifies an airline and specifies the source and destination airports of a route. The relevant fields are presented in Table 2.2.

Table 2.2 Relevant fields in the OpenFlights route dataset.

Field	Attribute	Description
1	Airline	Two-letter or other airline code.
2	Airline ID	Identifier assigned to the airline.
3	Source airport	Code of the departure airport.
4	Source airport ID	Identifier of the departure airport.
5	Destination airport	Code of the arrival airport.
6	Destination airport ID	Identifier of the arrival airport.
7	Codeshare	Indicates whether the route is operated under a codeshare arrangement.
8	Stops	Number of stops associated with the route record.
9	Equipment	Aircraft equipment codes.

The source and destination airport IDs are the main fields used to establish connections in the graph. In particular, the order of these two fields is preserved because a route from airport A to airport B does not necessarily imply the existence of a route from airport B to airport A.

2.2 Fundamentals of Graph Modeling

A graph is a mathematical structure used to represent relationships between objects. It is commonly defined as:

$$G = (V, E) \tag{2.1}$$

where V denotes the set of vertices (or nodes), and E denotes the set of edges connecting pairs of vertices.

For the airport route network, the graph elements are defined as follows:

- **Vertices (V):** Each vertex represents an airport. Airport IDs are used as unique node identifiers.
- **Edges (E):** Each edge represents a recorded direct route from a source airport to a destination airport.

Since flight routes have a defined direction, the airport network is modeled as a directed graph. A directed edge from node u to node v is written as (u, v), indicating a route from airport u to airport v.

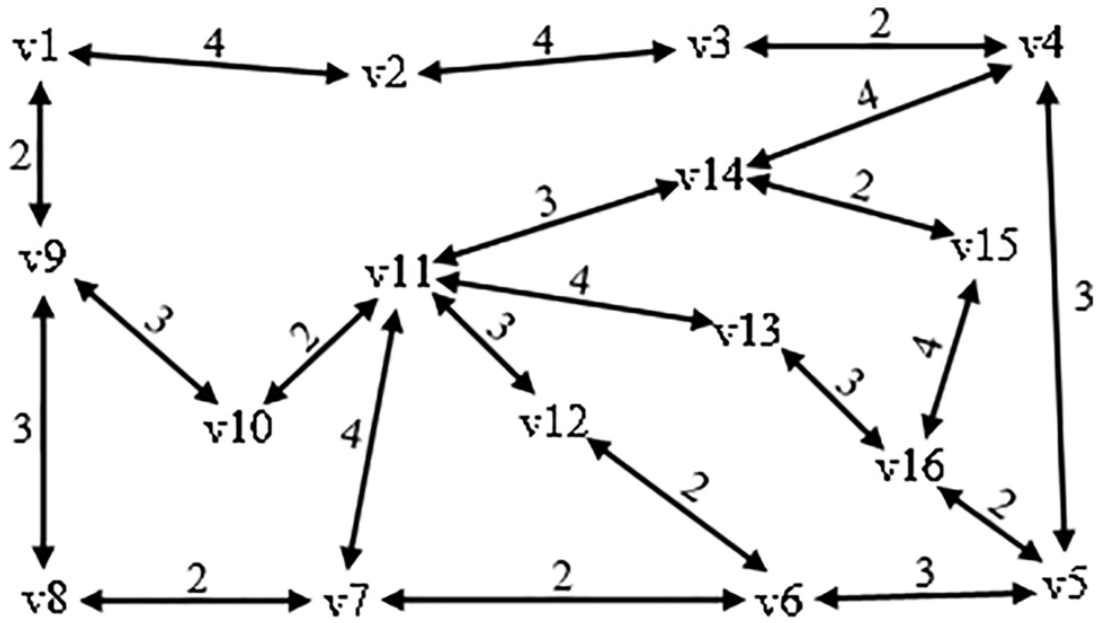


Figure 2.1 Directed graph representation of airport routes.

Figure 2.1 illustrates the conceptual representation of a directed airport route network, where airports are modeled as nodes and flight routes as directed edges. The edge weights shown in the figure are illustrative only and are not considered in this report.

In a directed network, the direction of an edge is significant. Consequently, the number of incoming connections and outgoing connections of an airport can be different. These quantities are known as the in-degree and out-degree, respectively.

2.3 Directed Graph and Directed Multigraph

The route dataset may contain multiple records connecting the same ordered pair of airports. For example, different airlines may operate routes between the same source and destination airports. Therefore, representing the network using only one edge per airport pair would discard some information contained in the original route records.

To address this issue, two graph representations are implemented using NetworkX: a directed multigraph and a directed graph.

Directed Multigraph

A directed multigraph allows multiple directed edges between the same pair of nodes. This representation is suitable for preserving individual route records, including parallel edges connecting the same source and destination airports.

In this project, the directed multigraph is implemented using NetworkX's `MultiDiGraph` class. Each valid route record is represented as a separate directed edge. This preserves the individual connections recorded in the original dataset.

Directed Graph

A directed graph allows at most one edge for each ordered pair of nodes. When multiple route records connect the same source airport to the same destination airport, they are represented by a single directed edge.

This representation is implemented using NetworkX's `DiGraph` class. It provides a simplified view of the network by focusing on the existence of directed connections rather than the number of individual route records.

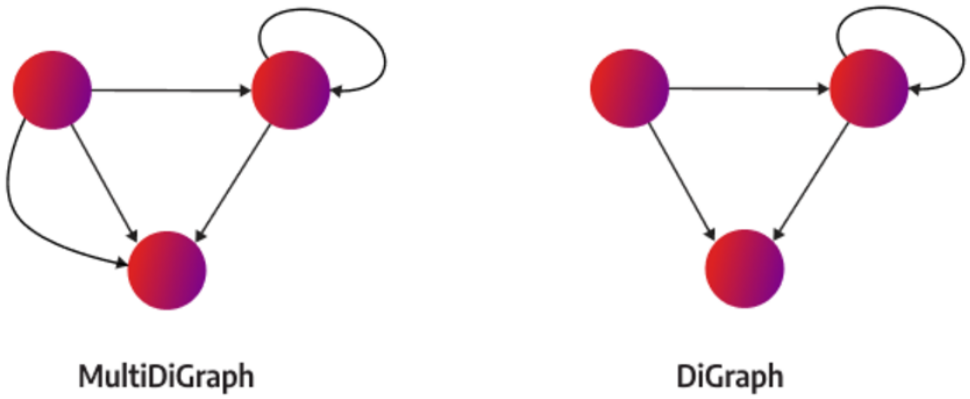


Figure 2.2 Comparison of directed multigraph and directed graph representations.

The two representations serve different analytical purposes. The directed multigraph preserves the route-level information available in the dataset, while the directed graph provides a simplified representation of unique directed airport connections. Both are used in the implementation to examine the network from these complementary perspectives.

2.4 Fundamental Network Measures

Several fundamental measures are used to describe the structure of the constructed airport network. These measures provide a quantitative basis for examining the connectivity of individual airports and the overall network.

In-degree and Out-degree

For a directed graph, the in-degree of a node is the number of edges entering that node, while the out-degree is the number of edges leaving it.

For a node v, these quantities are defined as:

$$k_{\text{in}}(v) = |\{u \in V : (u,v) \in E\}| \tag{2.2}$$

$$k_{\text{out}}(v) = |\{u \in V : (v,u) \in E\}| \tag{2.3}$$

The total degree of a node is the sum of its in-degree and out-degree:

$$k(v) = k_{\text{in}}(v) + k_{\text{out}}(v) \tag{2.4}$$

In the airport network, these measures describe the number of direct incoming and outgoing connections associated with each airport. The total degree is used to rank airports according to their number of direct connections.

Network Density

Network density measures the proportion of existing edges relative to the maximum number of possible edges.

For a directed graph without self-loops, the density is defined as:

$$D = \frac{|E|}{|V|(|V|-1)} \tag{2.5}$$

where $|V|$ is the number of nodes and $|E|$ is the number of directed edges.

A higher density indicates that a larger proportion of possible directed connections exists in the network. Conversely, a lower density indicates that the network contains relatively few connections compared with the maximum possible number.

Average Degree

The average in-degree and average out-degree describe the average number of incoming and outgoing connections per node.

For a directed graph, they are calculated as:

$$\langle k_{\text{in}} \rangle = \frac{|E|}{|V|} \tag{2.6}$$

$$\langle k_{\text{out}} \rangle = \frac{|E|}{|V|} \tag{2.7}$$

The average total degree is therefore:

$$\langle k \rangle = \frac{2|E|}{|V|} \tag{2.8}$$

These measures provide an overview of the connectivity of the network as a whole.

Degree Centrality

Degree centrality is a normalized measure of the connectivity of an individual node. For a directed graph, NetworkX calculates degree centrality using the total degree of a node:

$$C_D(v) = \frac{k(v)}{|V| - 1} \tag{2.9}$$

For directed graphs, this measure can theoretically reach a maximum value of 2 because a node may have incoming and outgoing connections to every other node.

In this project, degree centrality is used together with total degree to examine the connectivity of airports and identify those with the largest number of direct connections.

It is important to note that degree-based measures describe topological connectivity only. They do not directly represent passenger volume, flight frequency, or the operational capacity of an airport.

2.5 Summary

This chapter has introduced the OpenFlights dataset and the fundamental concepts used to model the global airport route network. Airports are represented as nodes, while recorded routes are represented as directed edges. Two graph structures, `MultiDiGraph` and `DiGraph`, are considered to preserve individual route records and represent unique directed connections, respectively.

The chapter has also introduced the basic network measures used in the subsequent analysis, including in-degree, out-degree, total degree, network density, average degree, and degree centrality. These concepts establish the theoretical and data-related foundation for the network construction and analysis procedures presented in the following chapters.

CHAPTER 3. NETWORK CONSTRUCTION AND DATA PROCESSING

3.1 Implementation Environment

The airport route network is constructed using Python and NetworkX, a library that provides data structures and algorithms for graph creation and analysis. NetworkX supports different types of graphs, including directed graphs and directed multigraphs, making it suitable for representing the characteristics of the OpenFlights route dataset.

Two graph objects are created during the implementation:

- `MultiDiGraph`: Preserves individual route records, including multiple edges between the same ordered pair of airports.
- `DiGraph`: Represents unique directed connections between airports, with repeated connections between the same ordered pair represented by a single edge.

The airport and route datasets are processed separately before being combined to construct the network. Airport identifiers are used to establish the correspondence between airport records and route endpoints.

3.2 Data Loading and Preprocessing

The first step is to load the two OpenFlights files: `airports.dat.txt` and `routes.dat.txt`. The airport file provides information about individual airports, while the route file contains the recorded connections between source and destination airports.

The initial dataset statistics obtained from the implementation are summarized in Table 3.1.

Table 3.1 Initial OpenFlights dataset statistics.

Dataset attribute	Count
Total airport records	7,698
Unique airport IDs	7,698
Total route records	67,663

During preprocessing, airport records are checked for valid geographical coordinates. The latitude and longitude values are important because they describe the geographical locations of airports and can be used in the subsequent network visualization.

In the loaded dataset, all 7,698 airport records have valid coordinates. The airport identifiers are also checked to ensure that they can be used as unique node identifiers.

For route records, the source and destination airport IDs are extracted. These identifiers are subsequently used to match each route endpoint with its corresponding airport record.

3.3 Airport Matching and Network Preparation

Not every airport listed in the airport dataset necessarily participates in a recorded route. Therefore, the airport records must be matched with the airport identifiers referenced in the route dataset before constructing the network.

The matching process consists of the following steps:

1. Extract the airport IDs referenced by the source and destination fields in the route dataset.
2. Compare these identifiers with the valid airport IDs obtained from the airport dataset.
3. Identify the airport records whose IDs are referenced by at least one route and have valid geographical coordinates.
4. Retain route records only when both the source and destination airport IDs can be matched with valid airport records.

Mathematically, let A denote the set of valid airport IDs obtained from the airport dataset, and let R denote the set of airport IDs referenced as either source or destination in the route dataset. The set of airports included in the network is therefore defined as:

$$V = A \cap R \tag{3.1}$$

where V contains only airports that have valid geographical coordinates and are referenced in the route dataset.

The matching results are summarized in Table 3.2.

After matching, 3,221 airports are available for network construction, together with 66,771 valid route records. Airport records that are not associated with the retained routes are not included in the initial network.

Table 3.2 Airport matching results.

Attribute	Count
Airports in airport dataset	7,698
Distinct airport IDs referenced in routes	3,334
Matched network airports	3,221
Route records retained after matching	66,771

3.4 Graph Construction Using NetworkX

After data preparation and airport matching, the network is constructed using the two graph representations introduced in Chapter 2.

Node Construction

Each matched airport is represented by a node in both graph structures. The airport ID is used as the node identifier to ensure that each airport has a unique representation.

Descriptive information from the airport dataset is retained as node attributes. These attributes include the airport name, city, country, IATA code, ICAO code, and geographical coordinates.

Using airport IDs rather than airport names or IATA codes avoids relying on descriptive fields as unique identifiers when establishing connections between airports.

Edge Construction

Each retained route record is represented as a directed edge from its source airport to its destination airport. The direction of the route is preserved throughout the construction process.

For the directed multigraph, every retained route record creates a separate edge. Consequently, multiple records between the same ordered pair of airports are preserved.

For the directed graph, repeated connections between the same source and destination airports are represented by a single directed edge. This produces a network that describes the existence of unique directed airport connections rather than the number of individual route records.

The resulting graph structures are summarized in Table 3.3.

The difference in edge counts results from the different ways in which the two graph structures handle repeated route records. The multigraph preserves individual records, whereas the directed graph represents each unique ordered airport pair with a single

Table 3.3 Network statistics before cleaning.

Network property	MultiDiGraph	DiGraph
Number of nodes	3,221	3,221
Number of edges	66,771	36,907
Self-loops	1	1
Isolated nodes	7	7

edge.

3.5 Network Validation and Cleaning

After graph construction, the network is examined to identify structural anomalies that may affect subsequent analysis. Two types of anomalies are considered: self-loops and isolated nodes.

Self-loop Detection

A self-loop is an edge that connects a node to itself. In the context of an airport network, this corresponds to a route record whose source and destination airport IDs are identical.

The implementation identifies one self-loop in the constructed network. The corresponding route record is shown in Table 3.4.

Table 3.4 Self-loop route record identified during validation.

Attribute	Value
Airline	IL
Airline ID	10121
Source airport	PKN
Source airport ID	3910
Destination airport	PKN
Destination airport ID	3910
Stops	0
Equipment	AT7

Since a route connecting an airport to itself does not represent a meaningful connection between two distinct airports in the scope of this study, this self-loop is removed from both graph structures.

Isolated Node Detection

An isolated node is a node that has no connections to any other node. In a directed graph, this means that the node has neither incoming nor outgoing edges.

The validation process identifies seven isolated airports in the constructed network. They are listed in Table 3.5.

Table 3.5 Isolated airports identified before cleaning.

Airport ID	Airport name	IATA	Country
7309	Charlotte Amalie Harbor Seaplane Base	SPB	Virgin Islands
3148	Rajahmundry Airport	RJA	India
7642	Tasiilaq Heliport	\N	Greenland
7217	Akiak Airport	AKI	United States
3151	Tirupati Airport	TIR	India
7235	Tok Junction Airport	TKJ	United States
6136	Lopez Island Airport	LPS	United States

These airports have no incoming or outgoing connections in the constructed network. Since the subsequent analysis focuses on the connectivity structure of the airport route network, isolated nodes are removed from both graph representations.

Cleaning Results

The cleaning process removes the identified self-loop and seven isolated nodes. The resulting network statistics are presented in Table 3.6.

Table 3.6 Network statistics after cleaning.

Network property	MultiDiGraph	DiGraph
Number of nodes	3,214	3,214
Number of edges	66,770	36,906
Remaining self-loops	0	0
Remaining isolated nodes	0	0

The cleaning results confirm that the final network contains no self-loops or isolated nodes. Removing the self-loop reduces the edge count by one in each graph, while removing the seven isolated nodes reduces the number of nodes without changing the edge count.

3.6 Network Export and Visualization Preparation

After cleaning and validation, the two graph structures are prepared for subsequent analysis and visualization. The cleaned network is exported so that its structural information can be reused without repeating the entire data preprocessing and graph construction procedure.

The exported network data provides the basis for the subsequent analysis of airport connectivity and the interactive visualization presented in the following chapters.

The geographical coordinates and descriptive attributes associated with airport nodes also provide the information required to associate network nodes with their corresponding geographical locations.

3.7 Summary

This chapter has described the implementation workflow used to construct the global airport route network from the OpenFlights dataset. The process includes loading the airport and route records, validating geographical coordinates, matching airport identifiers, and constructing two graph representations using NetworkX.

The initial network contains 3,221 nodes and 66,771 edges in the directed multigraph, while the directed graph contains 36,907 unique directed connections. Following the removal of one self-loop and seven isolated nodes, the final network consists of 3,214 airports, 66,770 route records in the multigraph, and 36,906 unique directed connections in the directed graph.

These cleaned graph structures provide the basis for the topological analysis and visualization presented in the next chapters.

CHAPTER 4. NETWORK ANALYSIS AND RESULTS

4.1 Overview of the Constructed Network

Following the data processing and cleaning procedures described in Chapter 3, the final airport route network consists of 3,214 airport nodes. Two graph representations are maintained: a directed multigraph containing individual route records and a directed graph representing unique directed connections between airports.

The main network statistics are summarized in Table 4.1.

Table 4.1 Overview of the cleaned airport route network.

Network property	MultiDiGraph	DiGraph
Number of nodes	3,214	3,214
Number of edges	66,770	36,906
Self-loops	0	0
Isolated nodes	0	0

The difference between the two edge counts reflects the presence of multiple route records between certain airport pairs. While the multigraph preserves these individual records, the directed graph represents each ordered airport pair with only one edge.

For the following topological analysis, the directed graph is selected because it provides a consistent representation of unique airport-to-airport connections. This avoids counting multiple route records between the same pair of airports as separate connections when examining network connectivity.

4.2 Network Density

Network density describes how many directed connections exist relative to the maximum number of possible directed connections in the network.

For the cleaned directed graph, the number of nodes is $|V| = 3,214$, while the number of edges is $|E| = 36,906$. Since the network contains no self-loops, its density is calculated as:

$$D = \frac{36,906}{3,214(3,214-1)} \tag{4.1}$$

Therefore,

$$D \approx 0.003574 \tag{4.2}$$

This corresponds to approximately 0.3574% of all possible directed connections.

The relatively low density indicates that only a small fraction of all possible airport-to-airport connections exists in the network. In other words, the airport network is sparse rather than fully connected.

This is consistent with the structure of an air transportation system, in which airports generally maintain connections with selected destinations rather than every other airport in the network.

4.3 Average Degree

The average in-degree and average out-degree describe the average number of incoming and outgoing connections per airport, respectively.

For a directed graph, both quantities are equal because every directed edge contributes one incoming connection and one outgoing connection.

Using the final network statistics:

$$\langle k_{\text{in}} \rangle = \frac{36,906}{3,214} \approx 11.4829 \tag{4.3}$$

Similarly,

$$\langle k_{\text{out}} \rangle = \frac{36,906}{3,214} \approx 11.4829 \tag{4.4}$$

The average total degree is therefore:

$$\langle k \rangle = \langle k_{\text{in}} \rangle + \langle k_{\text{out}} \rangle \approx 22.9658 \tag{4.5}$$

These results indicate that an airport has, on average, approximately 11.48 incoming and 11.48 outgoing unique directed connections in the constructed network.

It should be noted that these are network-wide averages. They do not imply that every airport has a similar number of connections. Some airports have substantially more connections than the average, as demonstrated by the degree ranking in the following section.

4.4 Airport Ranking Based on Total Degree

The total degree of an airport is calculated as the sum of its in-degree and out-degree. It provides a measure of the number of direct incoming and outgoing connections associated with that airport.

Using NetworkX, the total degree of each node is calculated and used to rank airports in descending order. The ten airports with the highest total degree are presented in Table 4.2.

Table 4.2 Top 10 airports ranked by total degree.

Rank	Airport	IATA	In-degree	Out-degree	Total degree
1	Frankfurt am Main Airport	FRA	238	239	477
2	Charles de Gaulle International Airport	CDG	233	237	470
3	Amsterdam Airport Schiphol	AMS	231	232	463
4	Atatürk International Airport	ISL	227	224	451
5	Hartsfield Jackson Atlanta International Airport	ATL	216	217	433
6	Chicago O'Hare International Airport	ORD	203	206	409
7	Beijing Capital International Airport	PEK	204	204	408
8	Munich Airport	MUC	189	191	380
9	Domodedovo International Airport	DME	188	187	375
10	Dallas Fort Worth International Airport	DFW	185	187	372

Frankfurt am Main Airport (FRA) ranks first, with a total degree of 477, consisting of 238 incoming and 239 outgoing connections. It is followed by Charles de Gaulle International Airport (CDG) with 470 connections and Amsterdam Airport Schiphol (AMS) with 463 connections.

The relatively high degree values of these airports indicate that they maintain a large number of direct connections within the constructed network. Several other major international airports, including Atlanta, Chicago, Beijing, Munich, and Dallas Fort Worth, also appear among the ten highest-ranked airports.

The results illustrate that the network's connections are not distributed uniformly among all airports. Certain airports have considerably more direct connections than the network-wide average, making them prominent nodes in terms of topological connectivity.

However, a high total degree should not be interpreted directly as evidence of higher pas-

senger traffic, greater flight frequency, or larger operational capacity. The ranking only reflects the number of unique incoming and outgoing airport connections represented in the dataset.

4.5 Degree Centrality Analysis

In addition to total degree, degree centrality is used to evaluate the relative connectivity of individual airports within the network.

For the directed graph, NetworkX calculates degree centrality by normalizing the total degree of each node with respect to the number of other nodes:

$$C_D(v) = \frac{k(v)}{|V| - 1} \tag{4.6}$$

Since the final network contains 3,214 nodes, the degree centrality of an airport is calculated by dividing its total degree by 3,213.

For example, the degree centrality of Frankfurt am Main Airport is:

$$C_D(\text{FRA}) = \frac{477}{3,213} \approx 0.14846 \tag{4.7}$$

The degree centrality values of the ten highest-ranked airports are presented in Table 4.3.

Table 4.3 Degree centrality of the top 10 airports.

Rank	IATA	Airport	Degree centrality
1	FRA	Frankfurt am Main Airport	0.14846
2	CDG	Charles de Gaulle International Airport	0.14628
3	AMS	Amsterdam Airport Schiphol	0.14410
4	ISL	Atatürk International Airport	0.14037
5	ATL	Hartsfield Jackson Atlanta International Airport	0.13477
6	ORD	Chicago O'Hare International Airport	0.12730
7	PEK	Beijing Capital International Airport	0.12698
8	MUC	Munich Airport	0.11827
9	DME	Domodedovo International Airport	0.11671
10	DFW	Dallas Fort Worth International Airport	0.11578

The ranking based on degree centrality is identical to the ranking based on total degree because all nodes are normalized by the same constant, $|V| - 1$. Consequently, airports with higher total degree also have higher degree centrality in this network.

For the airports listed in the table, the centrality values range from approximately 0.11578 to 0.14846. These values indicate that the top-ranked airports have a substantially larger number of direct connections than the average airport in the network.

Degree centrality therefore provides a normalized measure of airport connectivity, making it easier to compare nodes within the same network.

4.6 Discussion of Results

The analysis provides several observations about the topological structure of the constructed airport network.

First, the low network density indicates that the network is sparse. Although thousands of airports are represented, only a small proportion of all possible directed airport connections exists. This demonstrates that the network is not organized as a system in which every airport connects directly to every other airport.

Second, the average degree provides a general description of network connectivity, but the top-degree ranking shows that connections are not distributed uniformly. A relatively small group of airports has a much larger number of direct connections than the network-wide average. These airports can be considered highly connected nodes within the constructed network.

Third, the in-degree and out-degree values of the highest-ranked airports are generally close to each other. However, they are not necessarily identical. This reflects the directed nature of the airport network, in which incoming and outgoing connections are treated as distinct relationships.

Finally, the results demonstrate the usefulness of degree-based measures for identifying highly connected airports. Nevertheless, these measures describe only the direct connectivity structure. A more comprehensive assessment of airport importance would require additional information, such as passenger traffic, flight frequency, and airport capacity, which are not included in the current analysis.

4.7 Summary

This chapter has presented the topological analysis of the cleaned global airport route network. The analysis was performed using the directed graph containing 3,214 nodes and 36,906 unique directed connections.

The network density is approximately 0.003574, indicating a sparse network structure. The average in-degree and out-degree are both approximately 11.4829, while the average total degree is approximately 22.9658.

The degree ranking identifies Frankfurt am Main Airport as the most highly connected airport in the constructed network, followed by Charles de Gaulle International Airport and Amsterdam Airport Schiphol. The degree centrality results provide a normalized representation of the same connectivity ranking.

Overall, the results demonstrate how fundamental network measures can be applied to identify connectivity patterns and highly connected nodes in a real-world transportation network.

CHAPTER 5. NETWORK VISUALIZATION

5.1 Overview of the Visualization Approach

In addition to numerical analysis, graphical visualization provides an intuitive way to understand the structure of a complex network. While statistical measures describe the connectivity of the network quantitatively, visualization allows users to observe the geographical distribution of airports and the relationships between them.

For the global airport route network, geographical visualization is particularly useful because airports are distributed across different regions of the world. By associating each airport with its latitude and longitude, the constructed network can be presented on a geographical map, with airports represented by their locations and flight routes represented by connections between them.

The visualization developed in this project uses the cleaned airport network as its underlying data source. The geographical coordinates and network connections are used to present the airport route system in a geographical context.

The purpose of this visualization is to provide a complementary view of the network and make its structure more accessible, rather than replace the numerical analysis presented in Chapter 4.

5.2 Implementation Technologies

The visualization is implemented as a web-based interface using standard front-end technologies. The main components used in the implementation are described below.

HTML and CSS

HTML is used to define the structure and content of the web interface, while CSS is responsible for its visual presentation and layout.

These technologies provide the foundation for organizing the visualization components, including the map container and other interface elements.

Tailwind CSS

Tailwind CSS is used to support the styling and layout of the interface. Its utility-based approach allows visual elements to be arranged and styled consistently without requiring extensive custom CSS definitions.

It is used to establish the overall appearance of the web interface, including spacing, typography, colors, and component layout.

Leaflet

Leaflet is an open-source JavaScript library for interactive maps. It is used as the main mapping technology in this project to display the geographical distribution of airports and their associated route connections.

By using geographical coordinates from the airport dataset, the map can represent airport locations according to their actual positions. This provides a geographical perspective of the network that cannot be obtained from numerical network statistics alone.

Network Data Integration

The visualization uses the network data exported from the Python-based NetworkX implementation. This separates the network construction and analysis process from the front-end presentation layer.

The Python implementation is responsible for processing the original OpenFlights records, constructing the graph structures, and preparing the cleaned network data. The web interface uses the exported data to present the resulting network geographically.

This approach allows the network to be constructed and processed independently of the visualization interface.

5.3 Visualization Interface

The main visualization presents the airport network on a geographical map. Each airport is associated with its corresponding geographical coordinates, allowing its location to be represented on the map.

Connections between airports illustrate the relationships established by the route dataset. This geographical representation helps users observe how airports are connected across different regions and provides a more intuitive understanding of the network's spatial structure.

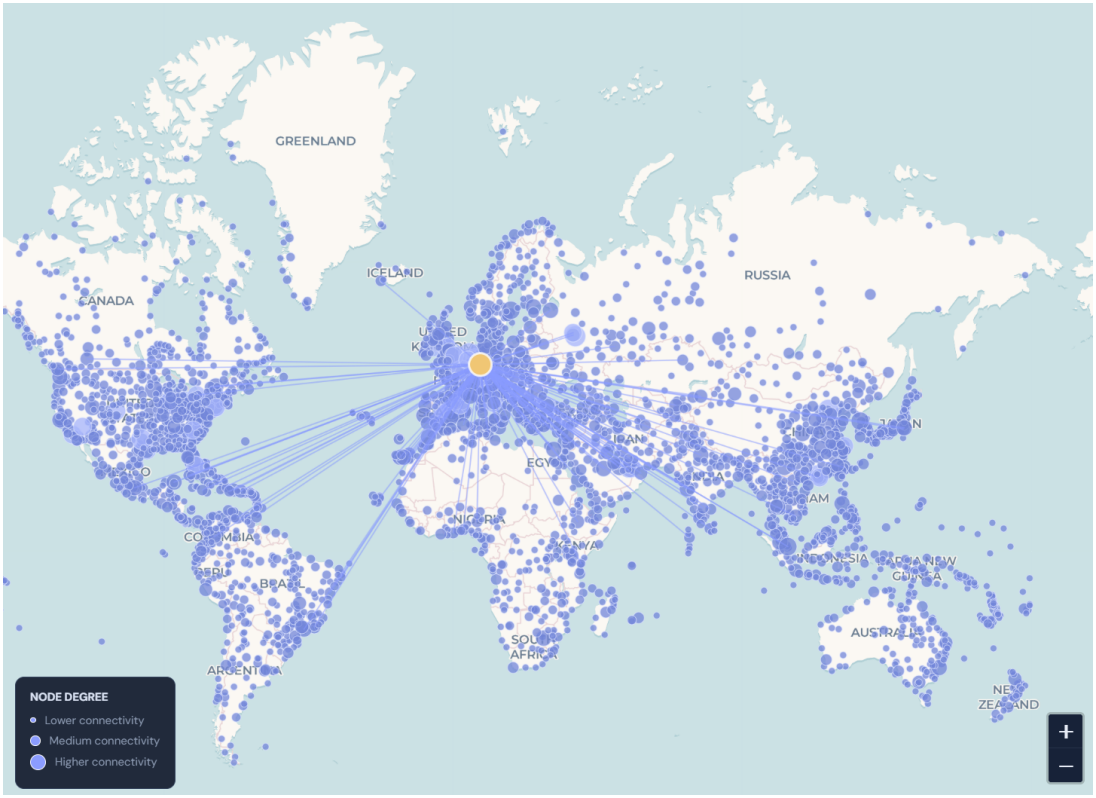


Figure 5.1 Geographical representation of the global airport route network.

The map provides a complementary perspective to the topological analysis. For example, the degree ranking in Chapter 4 identifies airports with a large number of direct connections, while the geographical visualization shows where these airports are located and how their connections are distributed spatially.

5.4 Deployment and Accessibility

The visualization is deployed as a web application, allowing users to access the interface through a web browser without having to execute the Python network construction script locally.

The separation between the Python implementation and the web interface also makes the project easier to maintain. The network can be processed and exported independently, while the front-end application is responsible for presenting the resulting data.

This deployment approach provides a convenient means of demonstrating the constructed network and sharing the results of the project.

5.5 Limitations of the Visualization

Although geographical visualization provides an intuitive representation of the airport network, some limitations should be considered.

First, the visualization depends on the completeness and accuracy of the underlying OpenFlights dataset. Since the dataset represents historical information last updated in June 2014, the displayed network does not necessarily reflect current airline routes or airport operations.

Second, a large number of connections can make the map visually dense, particularly in regions where many airports are located close to one another. This may make individual routes difficult to distinguish at certain zoom levels.

Finally, the geographical representation focuses on airport locations and recorded route connections. It does not directly communicate other operational characteristics, such as passenger volume, flight frequency, or airport capacity, because these attributes are not included in the implemented analysis.

Therefore, the visualization should be considered a graphical representation of the constructed network rather than a complete operational model of global air transportation.

5.6 Summary

This chapter has presented the web-based visualization developed to complement the construction and analysis of the global airport route network.

Using Leaflet for geographical mapping and front-end technologies for interface development, the visualization presents airport locations and their route connections on a world map. The exported network data connects the Python-based NetworkX implementation with the web interface.

The visualization provides an accessible way to explore the geographical structure of the network and complements the numerical results presented in Chapter 4.

CHAPTER 6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This report has presented the construction and analysis of a global airport route network using NetworkX and the OpenFlights dataset. The main objective was to demonstrate how existing real-world transportation data can be transformed into graph structures and analyzed using fundamental concepts of complex network theory.

The network was modeled using airports as nodes and recorded airline routes as directed edges. Two graph representations were implemented: a directed multigraph to preserve individual route records and parallel connections, and a directed graph to represent unique directed connections between airports. The resulting network was then processed to remove self-loops and isolated nodes before further analysis.

The cleaned network contains 3,214 airport nodes, 66,770 edges in the directed multigraph, and 36,906 unique directed connections in the directed graph. These graph structures provide the basis for examining the connectivity of the airport network.

Several fundamental network measures were calculated, including network density, average degree, and degree centrality. The directed graph has a density of approximately 0.003574 and average in-degree and out-degree values of approximately 11.48. The degree ranking also identifies Frankfurt am Main Airport, Charles de Gaulle International Airport, and Amsterdam Airport Schiphol as the three airports with the highest total degree in the constructed network.

These results illustrate that the airport network is sparse and that the number of direct network connections varies considerably among airports. They also demonstrate how graph-based measures can be used to identify highly connected airports and describe the overall connectivity structure of a real-world transportation network.

Finally, a web-based geographical visualization was developed to present the constructed network. By combining the NetworkX implementation with a map-based interface, the project provides both quantitative analysis and a geographical representation of the airport route system.

Overall, the project demonstrates the practical application of NetworkX in constructing, processing, visualizing, and analyzing a real-world complex network.

6.2 Limitations

Despite achieving the main objectives, the project has several limitations that should be considered when interpreting the results.

First, the analysis relies on the OpenFlights dataset, which represents historical airline route information last updated in June 2014. Consequently, the constructed network does not necessarily represent the current global airport network. Changes in airline operations, airport infrastructure, and route availability after that period are not reflected in the dataset.

Second, the analysis focuses primarily on the topological structure of the network. Each recorded route is represented as a connection between its source and destination airports, without considering the actual number of flights, passenger traffic, aircraft capacity, or other operational characteristics. Therefore, a high degree value indicates a large number of network connections, but does not necessarily imply higher passenger volume or operational importance.

Third, the current analysis is limited to basic network measures, including degree, density, and degree centrality. Other structural properties, such as clustering, shortest-path characteristics, community structure, and degree distribution, have not been investigated in this report.

Finally, the geographical visualization is subject to the limitations of the available dataset and the complexity of displaying a large number of connections on a single map. In densely connected regions, overlapping routes may reduce the visibility of individual connections.

6.3 Future Work

Several extensions could provide a more comprehensive understanding of the global airport network.

Extended Topological Analysis

Future work could investigate additional network properties, including degree distribution, clustering coefficient, shortest paths, and connected components. These measures would provide further insight into the structural organization of the network and the relationships between airports beyond their immediate connections.

Weighted Network Modeling

The current implementation treats connections between airports as unweighted edges. However, additional information such as flight frequency or passenger volume could be incorporated to construct weighted networks. This would allow the analysis to distinguish between connections with different levels of operational significance.

Temporal Network Analysis

Using datasets from multiple time periods could enable the investigation of how the airport network evolves over time. Such an approach could reveal changes in connectivity, the emergence of important hubs, and the effects of newly established or discontinued routes.

Further Investigation of Complex Network Models

The constructed airport network could also serve as a basis for studying theoretical models of complex networks. In particular, future work could investigate whether characteristics of the airport network are consistent with properties associated with scale-free networks and preferential attachment. This would provide a connection between empirical network analysis and the theoretical foundations of complex network science.

6.4 Final Remarks

Through this project, the process of transforming real-world data into graph representations has provided a practical introduction to complex network analysis. The use of NetworkX has demonstrated how graph structures can be constructed, processed, and examined through computational methods.

More importantly, the project establishes a foundation for further study of network science and its applications to real-world systems. The airport route network serves as an example of how relationships between individual entities can form a large-scale interconnected system whose structural properties can be investigated using graph theory.

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APPENDIX

Source Code Repository

The complete source code used in this report is publicly available on GitHub at:

`https://github.com/ndhminh/complex-networks-openflights`